Daniel Hu

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Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL
Frameworks & Data: PyTorch, Scikit-learn, Pandas, OpenCV, Streamlit, React, Angular, Node.js, PostgreSQL
Tools & Platforms: Git, Docker, Linux, AWS (S3, Bedrock, DynamoDB), ROS, Weights & Biases, Vercel, Jupyter
AI Development: Agentic System Design, LLM Orchestration, AI Coding Assistants (Claude Code, Cursor), MCP
Certifications: Career Essentials in Generative AI by Microsoft and LinkedIn; AWS DeepRacer Pro

Experience

Nationwide Columbus, OH
Software Engineering Intern | Nationwide Financial | Project Fulcrum *May 2026 – Present*

- Owned and built a business-critical module end to end (18 features, 22 endpoints) within an enterprise-wide financial platform, designing its PostgreSQL tables, building REST APIs, and an Angular/TypeScript UI
- Met frequently with the business team to gather requirements and verify the implementation matched their needs, translating requirements into structured specs that drove AI-assisted development with GitHub Copilot
- Contributing to a vendor replacement on pace to finish in 1 year and $1M after a prior 5-year, $6M failure
- Built a Black-Scholes pricing engine and Claude-powered competitive intelligence agent (AWS Bedrock, Streamlit) at the internal Enterprise Hackathon, cutting a 2 to 3 day actuarial pricing process to under a minute

Intelligent Design and Robotics Laboratory (OSU) Columbus, OH
Undergraduate AI/ML Researcher *Mar. 2024 – Aug. 2024; May 2025 – May 2026*

- Integrated a VLA policy via OpenCV and ROS for autonomous robot manipulation in real-world environments
- Trained CNN-based behavior cloning model to map wrist-camera input to 6-DOF UR5e arm actions
- Reproduced and extended Stanford's UMI environment; reengineered control logic to support Robotiq grippers
- Diagnosed and patched Python and C++ hardware drivers, resolving a critical 100% task failure rate
- Tracked and analyzed training runs with Weights & Biases, improving task completion rate by 62.5% overall

Big Data and Analytics Association Columbus, OH
AI Track Lead *Nov. 2025 – Present*

- Led the Foundations of AI track for a 300+ member org, overseeing teams building AI-powered products
- Mentored students on scoping, AI integration, and presentation; 10+ teams presented at the BDAA Research Gala

Projects

Disco | *JavaScript, Node.js, SQLite, Gemini API* | Repo | Demo 2026

- Built a social planning app where per-user AI agents turn friends' reviews into recommendations and outings
- Engineered agent-to-agent negotiation where agents trade only verdicts, never friends' budgets or schedules
- Grounded plans with Gemini and Google Search over 4,000+ Columbus venues, synced live across phones via SSE

Clairvoyant Loss Avoidance & Risk Advisor | *Python, IBM Granite LLM, LangChain* | Repo | Demo 2025

- Won 1st place in FinTech at IBM SkillsBuild × OSU Hackathon (100+ competitors) with an agentic risk platform
- Built a LangChain agent on IBM Granite that converts global news events into quantitative portfolio risk factors
- Monitors VaR/ES, triggers stop-loss alerts, and optimizes Black-Scholes hedges with a full audit trail

Sierra | *TypeScript, React, Node.js, Claude API, OpenAI, Mapbox* | Repo | Live 2025

- AI platform that plans Texas transmission line routes around ESA, CWA, NEPA, and NHPA compliance risk
- Ran multi-objective A* over a 560-node grid graph to return three ranked routes trading off cost and risk
- Orchestrated a Claude agent with 4 domain tools, MCP search, and regulatory RAG to estimate permit timelines

Education

The Ohio State University Columbus, OH
B.S. in Computer Science and Engineering | Statistics Minor | GPA: 3.8 *Aug. 2023 – Dec. 2026*
Engineering Honors Program

- Magna Cum Laude, Dean's List (All Semesters), National Merit Scholar, Study Abroad in Berlin (May 2024)
- Coursework: Artificial Intelligence, Software Development, Database Systems, Low-Level Programming, Data Structures & Algorithms, Operating Systems, Data Mining, Machine Learning, Agile Development, Statistics